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A Trajectory Data-Driven Personalized Autonomous Driving Decision System for Driving Simulators

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Abstract

To meet the high-fidelity testing environment requirements for autonomous driving system development, driving simulators are gradually evolving from tools that "only provide scenes and interaction interfaces" into integrated verification platforms for autonomous driving capabilities. These simulators, in particular, need to feature testable and scalable decision-making modules. However, the autonomous driving functions in existing driving simulators mostly rely on rule-based or simplified model approaches, which are inadequate for depicting the complex interactions in real-world traffic and fail to meet the personalized decision-making needs under various driving styles. To address these challenges, this paper designs and implements a trajectory data-driven personalized autonomous driving decision system, using drone aerial imagery as the core data source to provide realistic background traffic flow and human-like decision-making capabilities. The proposed system can be interpreted as an integrated decision–planning–control framework deployed within a high-fidelity driving simulation platform. It consists of a driving style classification module based on drone trajectory data, a personalized decision module integrating inverse reinforcement learning and dynamic game theory, and a planning and control module. First, a natural driving database is built using 4997 real vehicle trajectories, and prior features of different driving styles are extracted through trajectory feature engineering and an improved K-means++ method. Based on this, a personalized decision-making framework that combines dynamic game theory and maximum entropy inverse reinforcement learning is proposed, aiming to learn the preference weights of different driving styles in terms of safety, comfort, and efficiency. Furthermore, the Dueling Network Architecture (DuDQN) is used to generate human-like lane-changing strategies. Subsequently, a real-time closed-loop execution of personalized decisions in the simulation platform is achieved through fifth-order polynomial trajectory planning, lateral Linear Quadratic Regulator (LQR) control, and longitudinal cascade Proportional–Integral–Derivative (PID) control. Experimental results show that the personalized decision model trained with drone data can realistically reproduce vehicle decision-making behaviors in natural traffic flows within the simulation environment and generate autonomous driving strategies that are highly consistent with different driving styles. This significantly enhances the humanization and personalization capabilities of the autonomous driving module in the driving simulator.

Keywords: trajectory data; driving simulator; driving style recognition; inverse reinforcement learning; autonomous driving decision-making; trajectory planning and control

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Academic Editors: Zuoxian Gan, Linchao Li and Zhuangbin Shi

Received: 8 March 2026
Revised: 16 April 2026
Accepted: 17 April 2026
Published: 19 April 2026

1. Introduction

Intelligent connected vehicle technology has become a pivotal direction for the global automotive industry's development [1]. With the rapid advancement of Advanced Driver Assistance Systems (ADAS) and autonomous driving systems, the complexity of vehicles in perception, decision-making, and control continues to increase, imposing higher demands on algorithm verification and performance evaluation [2]. On one hand, real-world road testing is constrained by safety risks, operational condition repeatability, and testing costs, making it challenging to systematically cover extreme scenarios and diverse driving behaviors within a controlled environment [3]. On the other hand, pure software simulation still falls short in accurately reproducing traffic scene realism and traffic flow interactions, making it difficult to fully reflect the operational characteristics of autonomous driving systems in complex real-world traffic environments [4]. Therefore, constructing a novel experimental platform that combines high-fidelity driving environments with real-data-driven algorithm verification capabilities has become a key research direction in the field of intelligent vehicles [5].

In existing research, scholars both domestically and internationally have conducted extensive work on decision planning and testing verification for autonomous driving systems. Regarding testing platforms, vehicle-in-the-loop simulation, hardware-in-the-loop experiments, and indoor testing platforms based on driving simulators have gradually matured, significantly reducing the risks and costs associated with real-vehicle testing [6]. Oh et al. [7] proposed and validated a reliability assessment method for the Automated Driving Vehicle-in-the-Loop Simulation (AD-VILS) platform, aiming to enhance the accuracy and reliability of test results. Weiss et al. [8] developed a virtual reality-based vehicle-in-the-loop simulator that combines visual, vestibular, and haptic feedback to safely and in an immersive manner simulate highway-speed driving experiences within confined spaces. Fayazi et al. [9] developed a vehicle-in-the-loop testing environment to validate a traffic-light-free autonomous intersection control scheme and evaluate its performance in terms of traffic flow efficiency and fuel consumption. However, most platforms still rely on idealized or rule-based traffic flow models, with background vehicle behaviors lacking sufficient "human-like" characteristics and diversity. This makes it difficult to reflect the perturbations and challenges posed by different driving styles to autonomous decision-making [10].

The core challenge in autonomous decision planning lies in balancing safety, efficiency, and comfort while generating strategies consistent with human driving logic [11]. Current decision methods fall into three broad categories: rule-based approaches ensure basic safety but suffer from inflexible parameters that hinder adaptation to diverse driving styles [12]; learning-based methods capture behavioral patterns but lack interpretability [13]; hybrid approaches aiming to combine both strengths have recently garnered significant scholarly attention. For instance, Liu et al. [14] proposed a novel Curriculum Subgoal Inverse Reinforcement Learning (CSIRL) framework, guiding agent imitation learning by decomposing complex tasks into multiple local subgoals. CSIRL dynamically selects sub-objectives based on the agent's decision uncertainty when training on expert trajectories, then trains an intrinsic reward generator to construct local reward functions. Tian et al. [15] designed personalized parameters within an MPC-based lane-changing planning model, employing an apprentice learning approach to identify learnable parameters. They subsequently collected driver data to update these parameters online. Xu et al. [16] established a behavioral decision model based on lane priority, learning parameters for lane selection from demonstration datasets using maximum entropy IRL, and estimating acceleration parameters via particle filtering. While the aforementioned papers employed offline model parameter tuning, Likmeta et al. [17] proposed a direct interaction with the environment to adjust model parameters online: constructing a parameterized policy based on rules and updating parameters within the policy using Policy Gradients with Parameter-based Exploration.

With the advancement of drone and aerial fixed-camera technologies, aerial imagery data has emerged as a vital source for constructing natural traffic flow databases and analyzing interactive behaviors such as tailgating and lane changes. Its extensive coverage and strong spatiotemporal continuity enable a more comprehensive characterization of driving styles and behaviors [18]. Chouhan et al. utilized drone data to analyze driving behaviors in merging and diverging zones at toll stations within weakly lane-constrained traffic environments, quantifying characteristics and differences across multiple macro- and micro-level traffic parameters [19]. Xie et al. proposed a framework integrating driving style classification with lightweight width neural networks, employing drone aerial imagery to achieve high-precision analysis and prediction of human-like driving intent [20]. Domestic and international studies have employed in-vehicle or roadside sensors to collect natural driving data, utilizing clustering, machine learning, and rule-based methods for driving style recognition. However, most work remains confined to identification, with limited deep integration into autonomous driving decision-making [21]. Research applying UAV aerial data to driving simulator scenario reconstruction and establishing closed-loop validation mechanisms with autonomous driving systems remains relatively scarce [22].

To address the urgent need for high-fidelity, verifiable decision-making capabilities in autonomous driving development, this work constructs a drone data-driven personalized autonomous driving decision system for driving simulators, leveraging real drone aerial data and driving simulation technologies. The system integrates modules for driving-style modeling, personalized decision-making, trajectory planning, and control, and completes in-loop verification within a driving simulator. First, a relative-feature indicator system was established based on 4997 real vehicle trajectories, and an improved K-means++ method was applied to cluster driving styles, thereby deriving prior characteristics of different styles in terms of speed, headway, and interactive behaviors. On this basis, we propose a personalized decision-making framework that fuses dynamic game theory with maximum entropy inverse reinforcement learning to learn preference weights among safety, comfort, and efficiency from drone aerial demonstration data, and we combine this with DuDQN to generate human-like lane-changing strategies. Subsequently, quintic polynomial trajectory planning, lateral LQR control, and longitudinal cascade PID control are used to realize the real-time closed-loop execution of the policies on a high-fidelity driving simulation platform, followed by deployment and validation at the vehicle domain-controller level. Experimental results demonstrate that the system can realistically reproduce vehicle decision processes observed in natural traffic flows within the simulation environment and produce autonomous-driving behaviors that closely match different driving styles, offering a feasible, integrated technical pathway for drone data-driven personalized autonomous driving decision-making and its engineering integration into driving simulators.

2. Methodology

The drone data-driven personalized autonomous driving decision system is the core component enabling personalized behavior generation and closed-loop vehicle control in driving simulators. Its overall workflow is organized into three hierarchical layers: driving-style classification, personalized decision-making, and trajectory planning and control. First, using a natural-driving database constructed from drone aerial imagery, the system extracts inherent driver behavior patterns via feature engineering and clustering analysis, providing style priors for personalized policy learning. Next, by combining dynamic game theory with maximum entropy inverse reinforcement learning, an interpretable cost-function structure is constructed, and preference weights for safety, comfort, and efficiency are learned from expert demonstrations, thereby producing personalized decisions that are consistent with identified driving styles. Finally, quintic polynomial trajectory planning together with lateral LQR and longitudinal cascade PID control realizes real-time trajectory tracking and stable vehicle control, enabling accurate execution of decisions within the simulation environment. Through this tightly coupled, layered design, the platform generates autonomous driving behaviors that follow human driving logic, driven by real traffic behavior characteristics, and implements a complete perception–decision–planning–control closed loop. The overall platform architecture is shown in Figure 1. Arrows indicate data flow between modules. RLV and FLV denote the rear leading vehicle and front leading vehicle, respectively. Different colors represent different functional modules.

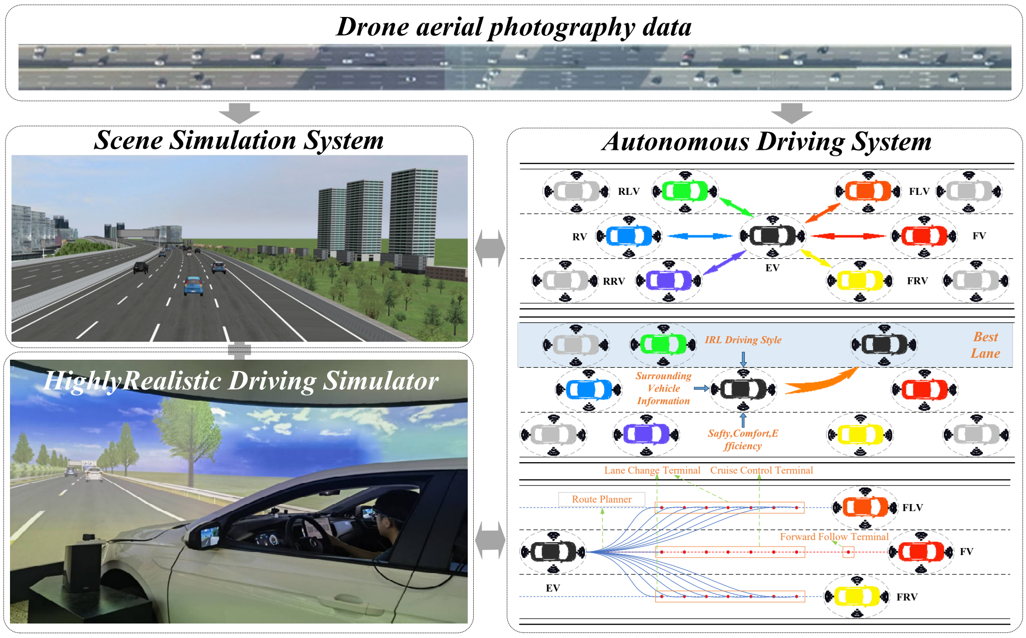

Figure 1. Platform Design Architecture Diagram.

2.1. *Drone–Aerial Data-Driven Driving-Style Classification Using Relative Characteristics*

To obtain authentic and natural driving behavior data, this study conducted long-term continuous observations of a typical section of Wuhan's urban expressway using a fixed aerial camera system. This section features frequent lane-changing behavior, providing abundant samples of straight driving and free lane changes. The data collection period was selected during the relatively stable afternoon traffic flow, accumulating approximately 3 h of high-resolution aerial video. This approach avoided interference from extreme congestion while ensuring sufficient vehicle interaction characteristics. Following image processing and trajectory extraction, a foundational natural driving dataset was constructed. This dataset includes attributes such as vehicle ID, timestamp, position, speed, acceleration, and lane number, comprising 4997 valid vehicle trajectories. It covers diverse traffic conditions ranging from free-flow to light congestion. Based on this dataset, lane-level traffic flow characteristics, vehicle interaction behavior features, and local neighborhood information of surrounding vehicles were extracted using the trajectory extraction method proposed by Liu et al. [23], providing data support for driving style modeling and background traffic flow construction. Figure 2 presents representative vehicle trajectories extracted from the drone aerial videos on the observed expressway section, illustrating typical longitudinal movement and lane-changing behaviors in the natural traffic flow.

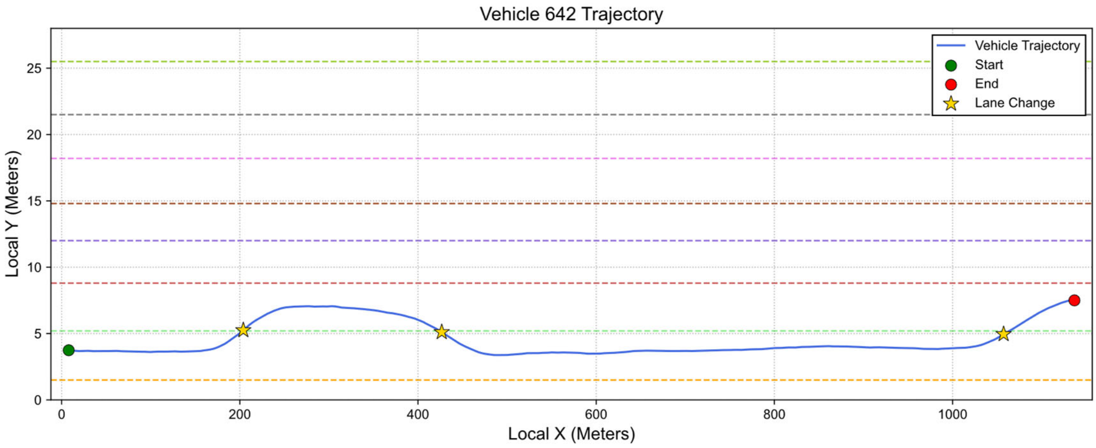

Figure 2. Representative vehicle trajectories extracted from drone aerial videos.

Driving style is a critical micro-level factor influencing traffic flow safety, comfort, and throughput efficiency. Its quantitative characterization forms the foundation for developing anthropomorphic and personalized autonomous driving strategies [24]. To this end, this study constructs a high-fidelity natural driving database using continuous trajectory data from urban expressways captured by fixed aerial cameras and proposes a systematic driving style recognition process. First, the raw trajectories undergo cleaning, calibration, and feature engineering to extract key behavioral parameters from three perspectives: following, lane changing, and vehicle interactions. Since absolute kinematic indicators are highly sensitive to traffic flow conditions and therefore cannot reliably reflect a driver's intrinsic long-term preferences, a comprehensive relative-feature-based metric system is established, as summarized in Table 1. Specifically, the following dimension is characterized by the relative relationships between the target vehicle and surrounding vehicles in terms of average speed and average headway time, reflecting the driver's speed preference, safety-margin preference, and risk acceptance during longitudinal car-following. The lane-changing dimension is characterized by the differences between the target vehicle and surrounding vehicles in lane-change frequency and lane-change gap characteristics, reflecting the driver's lane-change initiative, maneuverability, and operational aggressiveness. The vehicle-interaction dimension is characterized by the differences between the target vehicle's overtaking and being-overtaken behaviors and the average levels of surrounding vehicles, reflecting the driver's interaction intensity, competitive tendency, and behavioral dominance in the traffic flow.

By comparing individual feature values with the average feature values of the neighboring vehicle group within the same spatiotemporal range, this metric system effectively reduces the interference caused by heterogeneous traffic environments and enables more accurate extraction of inherent driver behavior patterns. The corresponding metric definitions and calculation formulas are summarized in Table 1.

Based on the clustering results, the 4994 valid trajectories were classified into three driving-style categories: cautious, normal, and aggressive. Specifically, 2127 trajectories (42.6%) were identified as cautious, 1653 trajectories (33.1%) as normal, and 1214 trajectories (24.3%) as aggressive. These three categories were defined according to the relative behavioral characteristics summarized in Table 1 and the clustering outcomes of the improved K-means++ algorithm, providing the basis for subsequent analyses of style-dependent decision-making behavior.

Table 1. Inherent Driving Style Metric System.

Characteristics	Composition	Calculation
Relative to the Chasing Style Characteristics	a: Average Speed of Ego Vehicle (AS) b: Average Speed of Ego Vehicle Relative to Surrounding Vehicles (GAS) c: Average Headway Time of Ego Vehicle (ATHW) d: Average Headway Time of Ego Vehicle Relative to Surrounding Vehicles (GATHW)	$a/b - c/d$
Relative Lane Change Style Characteristics	a: Number of lane changes by Ego vehicle (NLC) b: Average number of lane changes by surrounding vehicles (ANLC) c: Headway distance during lane changes by Ego vehicle (LCSHW) d: Minimum headway distance during lane changes between Ego vehicle and surrounding vehicles (MLCSHW)	
Relative Interaction Style Characteristics	a: Number of times Ego vehicle has overtaken (OF) b: Average number of overtakes by surrounding vehicles (AOF) c: Number of times Ego vehicle has been overtaken (BOF) d: Average number of overtakes by Ego vehicle relative to surrounding vehicles (ABOF)	

Relative to the Chasing Style Characteristics: This feature primarily characterizes the driver's tendency to maintain a following state within traffic flow. Its calculation is based on the relative difference between the vehicle's average speed, average headway, and the average values of the surrounding vehicle group. The specific calculation formula is:

$$AS = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} v_i(t)dt \;\; and \;\; GAS = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} v_i(t)dt \tag{1}$$

$$ATHW = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} h_i(t)dt \;\; and \;\; GATHW = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} h_i(t)dt \tag{2}$$

The calculation methods for features AS and GAS are shown in Equation (1). Where, t_1 and t_2 represent the temporal start and end points of vehicle i's trajectory, respectively; v_i denotes the instantaneous vehicle speed; and n represents the six vehicles surrounding vehicle i, fixed at 7. Offline simulations during the calibration phase demonstrated that expanding the neighborhood beyond six vehicles exponentially increases state-space complexity with negligible impact on the final decision outcome.

Similarly, the calculation methods for features ATHW and GATHW are shown in Equation (4), where h_i denotes the instantaneous headway time distance of vehicle i. A higher relative following style value indicates that the driver tends to follow at a faster speed and with a smaller distance, exhibiting more aggressive following characteristics.

Relative Lane Change Style Characteristics: This feature quantifies the frequency and risk-taking tendency of a driver's lane-changing behavior. It is reflected by comparing the vehicle's lane-change frequency and the front–end distance during lane changes against the average levels of surrounding vehicles. The specific calculation formula is:

$$NLC = \sum_{t=t_1}^{t_2} \delta(t) \ \ and \ \ ANLC = \frac{1}{n}\sum_{i=1}^{n} \delta_i(t_1, t_2) \tag{3}$$

$$LCSHW_i = \frac{1}{NLC(t_1, t_2)} \sum_{k=1}^{NLC(t_1,t_2)} \omega_k \ \ and \ \ MLCSHW = \min_{i=1,\ldots,n} \{LCSHW_i\} \tag{4}$$

The calculation methods for features NLC and $ANLC$ are shown in Equation (3), where δ represents the indicator function for lane change occurrence by vehicle i at time t. $\delta(t) = 1$ indicates a lane change occurred, otherwise $\delta(t) = 0$. The calculation methods for features $LCSHW$ and $MLCSHW$ are shown in Equation (4), where ω represents the front–end distance of vehicle i during its kth lane change. The lane change time refers to the moment when the lane ID changes. A higher relative lane-change style characteristic value indicates that the driver changes lanes more frequently and demands stricter spatial gaps during lane changes, reflecting a more aggressive driving style.

Relative Interaction Style Characteristics: This feature focuses on overtaking interactions between vehicles, reflecting drivers' competitive tendencies within traffic flow. It integrates comparisons of the vehicle's overtaking frequency and being overtaken frequency against the group average. The specific calculation formula is:

$$OF = \sum_{t=t_1}^{t_2} \lambda(t) \ \ and \ \ AOF = \frac{1}{n}\sum_{i=1}^{n} \lambda_i(t_1, t_2) \tag{5}$$

$$BOF = \sum_{t=t_1}^{t_2} \mu(t) \ \ and \ \ ABOF = \frac{1}{n}\sum_{i=1}^{n} \mu_i(t_1, t_2) \tag{6}$$

The calculation methods for feature OF and AOF are shown in Equation (5), where λ represents the indicator function for vehicle i overtaking at time t. $\lambda(t) = 1$ indicates an overtaking event occurred, otherwise $\lambda(t) = 0$. Similarly, the calculation methods for features BOF and $ABOF$ are shown in Equation (6), where μ represents the indicator function for vehicle i being overtaken at time t, with $\mu(t) = 1$ indicating overtaking and $\mu(t) = 0$ otherwise. A higher relative interaction style feature value indicates that the driver engages in proactive overtaking far more frequently than passive overtaking, occupying a dominant competitive position within the traffic flow.

Clearly, the underlying concept in constructing the above indicator system is to calculate the ratio of a specific characteristic value to the average value of all vehicles within the same spatiotemporal range. This approach aims to eliminate variations in driving performance caused by differing traffic conditions. This metric system condenses each driver's complex behavior into three core indicators. Together, they form the input feature vector for the driving style clustering model. The K-Means clustering algorithm is then applied to analyze these clusters. The resulting driving styles and travel characteristics are illustrated in Figure 3.

As shown in Figure 3, the aggressive driving style exhibits the highest speed, while the conservative driving style has the lowest speed, indicating that the aggressive style places the highest priority on travel efficiency. The aggressive driving style also demonstrates the greatest acceleration, followed by the conservative style, with the moderate style showing the least acceleration. This suggests that the moderate driving style prioritizes comfort more than the other styles. The headway distances for conservative, normal, and aggressive driving styles decrease in that order, indicating a corresponding reduction in safety emphasis among the three.

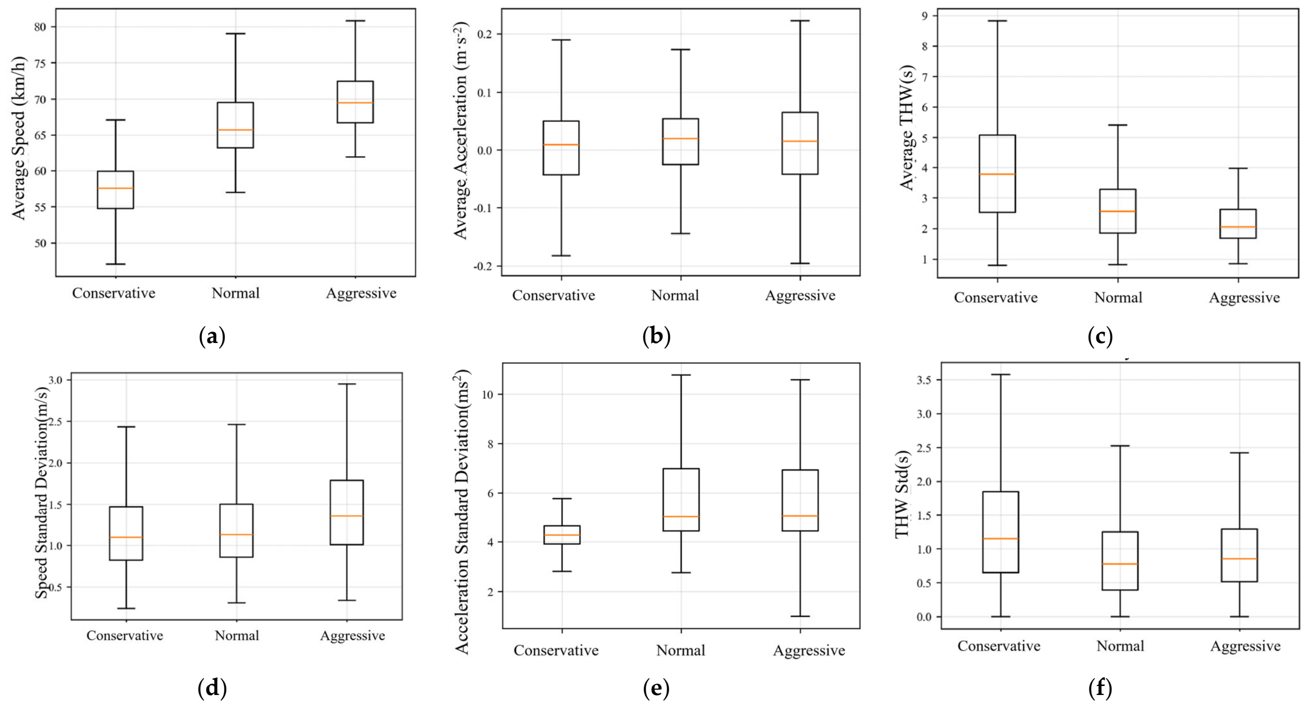

Figure 3. Driving Style Classification Features. (**a**) Distribution of Speed Average, (**b**) Distribution of Acceleration Averages, (**c**) Distribution of Average Headway, (**d**) Distribution of Speed Standard Deviations, (**e**) Distribution of Acceleration Standard Deviations, (**f**) Distribution of Headway Standard Deviations.

2.2. Personalized Decision System Integrating Inverse Reinforcement Learning and Dynamic Game Theory

This study constructs a personalized decision-making framework integrating dynamic game theory with Maximum Entropy Inverse Reinforcement Learning [25]. First, a parameterized cost function model is established based on a three-vehicle game relationship, explicitly embedding safety costs, comfort costs, and efficiency costs into the game structure to form interpretable decision constraints. Subsequently, Maximum Entropy IRL is employed to learn optimal weightings for the cost function from the behaviors of lane-changing experts with diverse driving styles, enabling the mapping from abstract styles to quantifiable preferences. Based on the learned personalized cost function, the DuDQN algorithm is employed to solve for the optimal policy. This enables the model to generate human-like decisions aligned with diverse driving styles in complex interactive scenarios. Its behavioral consistency and interpretability are validated through comparison with real aerial trajectory data. Vehicle inputs at each time step are defined as follows:

$$I_i^{(t)} = \left[trajectory_i^{(t)}, interact_i^{(t)}\right] \quad (t = t - T_p, \ldots, t-1, t) \quad (i = 0, 1, \ldots, 6) \tag{7}$$

where i denotes the identifier of a surrounding vehicle, 0 represents the current vehicle, and 1–6 respectively indicate the six closest vehicles relative to the current vehicle in the following directions: front-left, front, front-right, rear-left, rear, and rear-right. Vehicle historical trajectory $trajectory_i^{(t)} = \{x_i, y_i, v_i, a_i\}$ includes lateral position, longitudinal position, absolute velocity, and absolute acceleration. Interaction information $interact_i^{(t)} = \left\{\Delta x_i^{(t)}, \Delta y_i^{(t)}, \Delta v_i^{(t)}\right\}$ indicates the lateral relative distance, longitudinal relative distance, and relative velocity of vehicle i relative to the target vehicle at time t.

To provide a learning framework with clear physical meaning and interpretability for inverse reinforcement learning, this study constructs a cost function framework

https://doi.org/10.3390/vehicles8040094

for lane-changing decisions based on dynamic game theory. In the context of free lane changes on expressways, a single-step static game is constructed with the minimal participant set comprising "own vehicle–rear vehicle in target lane–front vehicle in target lane," where $a = \{left,\ right,\ keep\}$. The lane change decision function for the own vehicle is defined as:

$$R(s,a) = \omega_{sa}c_{sa} + \omega_{com}c_{com} + \omega_{ef}c_{ef} \tag{8}$$

The state $s = (v_x,\ v_y,\ a_x,\ a_y,\ \Delta x_i,\ \Delta v_i,\ l)$, where $v_x,\ v_y,\ a_x,\ a_y$ denote the vehicle's longitudinal and lateral velocity and acceleration, and $\Delta x_i,\ \Delta v_i$ represent the relative longitudinal distance and velocity between the vehicle and surrounding vehicles, with i indicating the identifier of the surrounding vehicle. Equations (1)–(6) denote the six closest vehicles relative to the vehicle's front–left, front, front–right, rear–left, rear, and rear–right directions, respectively. The vehicle's action set $a \in \{left,\ right,\ keep\}$ represents lane change to the left, lane change to the right, and maintaining the current lane, respectively. Window W: A historical frame window anchored to the lane-change event frame, used for constructing state features and cost evaluation. Its range is 0 to T_p where T_p denotes the historical time domain.

The lane-change decision cost function comprises three components: driving safety c_{sa}, passenger comfort c_{com}, and travel efficiency c_{ef}. The integrated cost function assigns distinct weighting coefficients to each cost factor $\omega_{sa}, \omega_{com}, \omega_{ef}$—to accommodate the personalized needs of autonomous vehicle passengers, thereby simulating diverse driving styles.

Safety Cost (c_{sa}): Integrating longitudinal approach risk and lateral lane-change TTC risk, the vehicle's safety cost c_{sa} increases as the relative speed between vehicles approaches and the front/rear vehicle spacing in the target lane decreases. It can be expressed as:

$$c_{sa}(s,a) = k_{THW}\left(\frac{THW_{ref} - THW}{THW_{ref}}\right) + k_{TTC}\left(\frac{TTC_{ref} - TTC(a)}{TTC_{ref}}\right) + k_c I_{Publish}(a) \tag{9}$$

THW and THW_{ref} represent the headway distances between the current vehicle and the preceding vehicle on the reference trajectory, characterizing longitudinal approach risk. TTC and TTC_{ref} denote the time-to-collision intervals between the current vehicle and the following vehicle in the target lane on the reference trajectory. IPublish is an indicator function whose value is either 1 or 0. If a collision risk is predicted within the future time window W based on the current decision action a, the value increases with greater risk. This treats potential collisions as a hard constraint to ensure safety.

Comfort Cost (C_{com}): The square integral of the second/third derivative based on a fifth-order fit of the windowed lateral trajectory, reflecting lateral acceleration/acceleration load. For the trajectory $x(t),\ y(t)$ within the time window W corresponding to action a, where $t \in [0,\ T_p]$, discretized into N segments with $\Delta t = T_p/N$, the comfort cost ccom can be expressed as:

$$c_{com} = \int \left[\lambda_{ax}a_x^2(t) + \lambda_{ay}a_y^2(t) + \lambda_{jx}j_x^2(t) + \lambda_{jy}j_y^2(t)\right]dt \tag{10}$$

where, a_x and a_y, j_x and j_y denote the vehicle's acceleration and jerk, respectively. λ^* represents the corresponding weighting coefficient used to accentuate the abrupt change in the lateral coordinate during lane changes.

Efficiency Cost (c_{ef}): The quadratic term of deviation from the reference speed, incentivizing efficiency while constrained by safety and comfort: Traffic efficiency primarily relates to cruising speed and following distance. Therefore, the efficiency cost c_{ef} can be expressed as:

$$c_{ef} = (\frac{v_{ref} - v_{aim}}{v_{ref}})^2 \quad (11)$$

where v_{aim} denotes the achievable velocity after executing action a, while vref represents the velocity of the expert trajectory.

Next, we perform inverse reinforcement learning based on expert trajectories: First, anchor a 10-frame window using the lane-change event frame, with negative samples sourced from long-duration holding intervals. Preliminary tests indicate that a 10-frame window provides an optimal balance between intention prediction accuracy and real-time computational efficiency. The window must satisfy conditions of small lateral displacement and velocity continuity, extract trajectory data for the target vehicle and surrounding vehicles within the window, and then compute game-theoretic costs c_{sa}, c_{com}, and c_{ef}, using these three cost types as features. We assume expert behavior follows an optimal strategy generated according to the maximum entropy principle. Maximum Entropy IRL learns the reward function by maximizing the log-likelihood of the expert trajectory:

$$p(\tau|\omega) = \frac{1}{Z(\omega)} exp(\sum_{t=0}^{T_p - 1} R(s_t, a_t)) \quad (12)$$

$$L(\omega) = \sum_{\tau} [\sum_{t} R(s_t, a_t) - log Z(\omega)] - \frac{\lambda}{2} \|\omega\|_2^2 \quad (13)$$

where, s_t denotes the current state, τ represents the expert trajectory, $Z(\omega)$ is the partition function used to normalize trajectories, and λ is the regularization term preventing overfitting. The log-likelihood function $L(\omega)$ estimates the optimal reward function by maximizing the probability of the expert behavior trajectory.

IRL learns the optimal reward function such that the expert's trajectory probability is maximized under the given reward function. It is worth noting that the weights within the functions are dynamically optimized through the IRL process. The algorithmic convergence naturally acts as an implicit sensitivity evaluation, ensuring that the selected parameters are robust and closely mirror human-like driving preferences, without the need for manual, heuristic tuning.

This study employs inverse reinforcement learning based on three expert trajectories classified from aerial survey data to obtain the following cost weights, as shown in Table 1. Training result statistics are presented in Table 2:

Table 2. Weight Table for Inverse Reinforcement Learning.

Factor	Aggressive	Normal	Conservative
ω_{sa}	0.14	0.46	0.65
ω_{com}	0.09	0.31	0.21
ω_{ef}	0.77	0.23	0.14

Once the reward function $R(s, a)$ is obtained, we use the DuDQN algorithm [26] to solve for the optimal policy. At each time step, the system calculates relevant features based on the current state st and estimates the Q-value through a neural network. Let $Q_{behave}(s_t,\ a_t)$ denote the Q-value corresponding to state st and action at. At each time step, we select an action using the behavior Q-network. Selection follows an epsilon-greedy policy: with probability ϵ, we choose a random action for exploration; with probability $1 - \epsilon$, we select the action with the highest current Q-value:

$$a_t = \begin{cases} \text{arg}max_a Q_{behave}(s_t, a; \theta) & 1 - \epsilon \\ \{left, keep, right\} & \epsilon \end{cases} \tag{14}$$

where ϵ represents the exploration rate, which gradually decreases as training progresses to encourage exploitation over exploration. The Q-values are updated using the dual DQN update rule. The target Q-values are computed and updated by the target network:

$$y_t = r_t + \gamma Q_{target}(s_{t+1}, argmax_a Q_{behave}(s_{t+1}, a)) \tag{15}$$

$$L_{DQN} = E\left[\left(Q_{behave}(s_t, a_t) - y_t\right)^2\right] \tag{16}$$

Once the Q-network training is complete, we can generate a policy by selecting the action corresponding to the maximum Q-value:

$$a_{opt}(s) = argmax_a Q_{behave}(s, a) \tag{17}$$

By employing the DuDQN algorithm, we can efficiently learn optimal lane-change decision strategies within complex state spaces while ensuring these decisions align with diverse driving styles. The following table compares the generated strategies. After iterative model convergence, the decision strategy's accuracy relative to expert decisions is shown in Table 3:

Table 3. Decision-Making Accuracy of Reinforcement Learning for Different Driving Styles.

Factor	Left	Right	Keep	Total
Aggressive	0.86	0.87	0.97	0.90
Normal	0.87	0.83	0.84	0.85
Conservative	0.83	0.91	0.87	0.87

It is evident that the sample distribution across the three action categories is significantly imbalanced. Accordingly, beyond reporting overall accuracy, this study further introduces the following complementary metrics to achieve a more comprehensive evaluation of decision performance:

(1) Class-wise precision and recall: special emphasis is placed on the recall of lane-change actions, which indicates whether the model is capable of correctly identifying the actual timing of lane-change decisions.
(2) Lane-change F1-score: as the harmonic mean of precision and recall, the F1-score is employed to alleviate the influence of class imbalance and to provide a more balanced assessment of lane-change recognition performance.

The three strategies share the same network architecture and hyperparameter configuration. Each strategy was trained for 200 episodes under five random seeds, and the results are presented in the form of mean ± standard deviation. The training convergence curves are illustrated below.

Figure 4 compares the training rewards under the three driving styles. All curves quickly enter a stable oscillation regime in the early stage, indicating that the policies can converge under the current settings. The aggressive style achieves the highest cumulative reward, followed by the normal style, while the cautious style yields the lowest reward. This ranking remains stable across different initializations. In addition, the training and validation curves are generally consistent, with no persistent divergence observed, suggesting good generalization consistency of the learned policies.

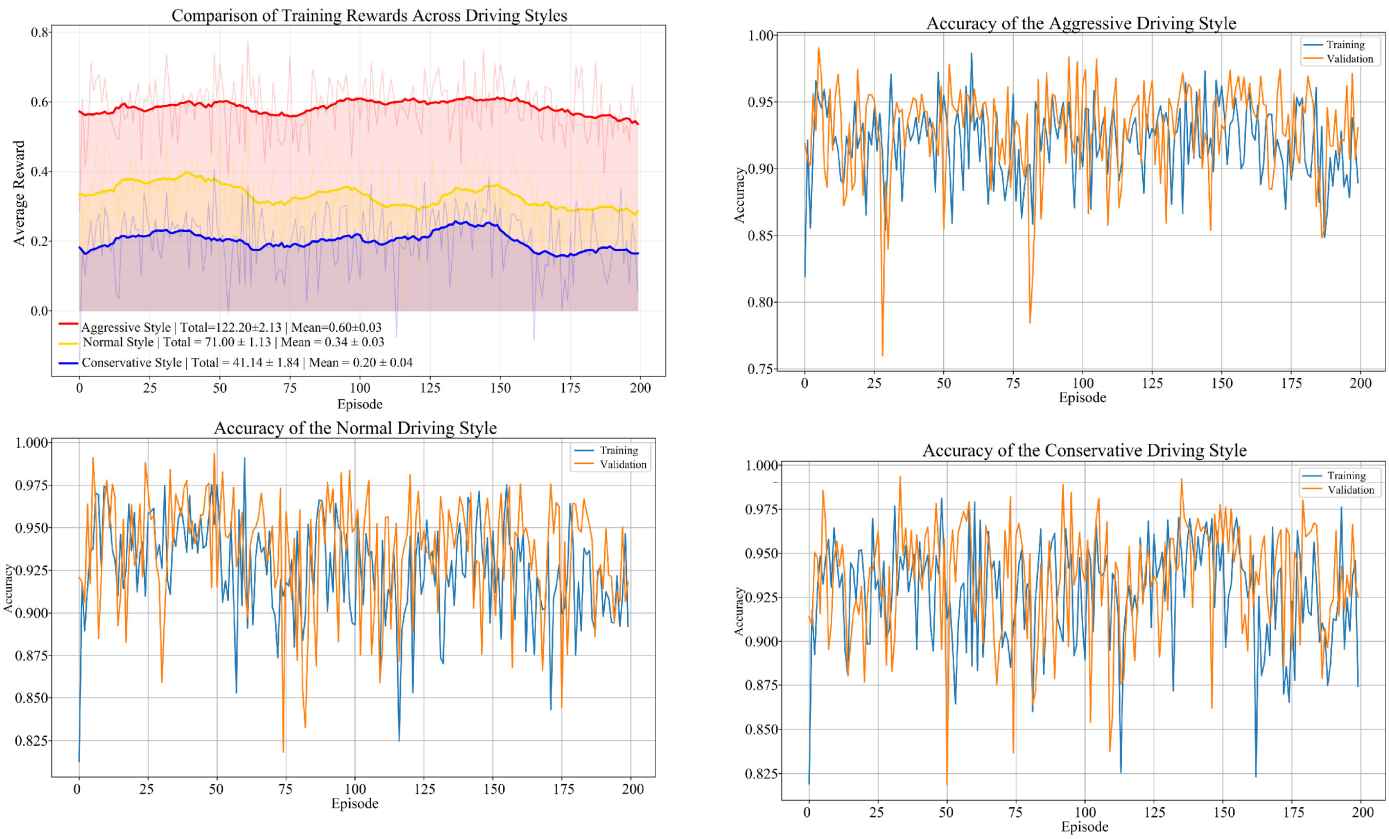

Figure 4. Training Reward and Accuracy Curve.

Table 4 summarizes the decision-matching metrics of policies with different driving styles. In terms of overall accuracy, the cautious style achieves the highest value; however, this is largely attributable to the high proportion of keep actions. After introducing the lane-change F1-score, it can be observed that the three styles exhibit comparable lane-change decision-making capability. Regarding lane-change recall, the aggressive style achieves the highest score, indicating that the model is better able to capture the lane-change decisions of aggressive drivers. By contrast, the cautious style shows a relatively lower recall, mainly because cautious drivers tend to have shorter and less salient lane-change decision windows, causing some lane-change maneuvers to be classified as keep actions during the early preparation stage.

Table 4. Decision-Matching Metrics of Different Style Policies.

Style	Overall Accuracy	Lane-Change Accuracy	Lane-Change Recall	Lane-Change F1-Score	Keep Accuracy
Aggressive	0.87 ± 0.04	0.74 ± 0.03	0.81 ± 0.02	0.80 ± 0.02	0.90 ± 0.01
Normal	0.90 ± 0.03	0.77 ± 0.02	0.78 ± 0.03	0.77 ± 0.02	0.93 ± 0.01
Conservative	0.94 ± 0.04	0.79 ± 0.04	0.72 ± 0.03	0.75 ± 0.03	0.97 ± 0.01

To verify the superiority of the proposed strategy, this study compares it with a traditional rule-based game model and a pure DDQN model. The results are presented in Table 5. The rule-based model obtains the lowest F1-scores, indicating that fixed-rule methods have difficulty capturing the real decision logic of drivers under multi-factor coupling. This limitation is especially evident for aggressive drivers, whose behavior is more context-dependent and flexible, and therefore cannot be adequately described by parameterized rules. Although the pure DDQN model achieves F1-scores close to those of the proposed method, its unsafe decision rate is significantly higher, which verifies the critical role of the game-theoretic safety safeguard in improving safety. The proposed method achieves the best F1-scores while maintaining the lowest unsafe decision rate, thereby demonstrating the necessity and effectiveness of combining IRL-based weight learning, DDQN-based policy generation, and rule-constrained safety fallback.

Table 5. Reinforcement Learning Training Results.

Method	Aggressive F1	Normal F1	Cautious F1	Unsafe Decision Rate
Rule-based model	0.58	0.62	0.64	2.1%
Pure DDQN	0.68	0.70	0.71	4.8%
Proposed method	0.80	0.77	0.75	1.4%

To verify whether the learned policies exhibit interpretable macroscopic style differences, Table 6 reports multi-seed statistics in terms of average gap (safety), lane-change frequency (comfort), average speed (efficiency), and final reward (return). The results show a clear monotonic pattern: the aggressive style has the smallest gap, the highest lane-change frequency, the highest speed, and the highest final reward; the cautious style has the largest gap, the lowest lane-change frequency, the lowest speed, and the lowest final reward; and the normal style remains at an intermediate level for all these indicators. The style differences in average gap and final reward are significantly larger than the corresponding standard deviations, indicating strong discriminability in key safety and return-related metrics. In contrast, lane-change frequency and speed exhibit relatively larger fluctuations, suggesting that traffic-segment variations and random initialization introduce some disturbance, although the overall trend remains consistent.

Table 6. Style Differentiation Results Under Reinforcement Learning.

Style	Average Gap/m	Lane-Change Frequency/%	Average Speed/m/s	Final Reward
Aggressive	13.23 ± 0.02	$21.88 \pm 4.33\%$	23.17 ± 2.13	0.59 ± 0.03
Normal	16.75 ± 0.04	$14.65 \pm 2.72\%$	21.81 ± 1.84	0.31 ± 0.03
Conservative	21.37 ± 0.04	$10.97 \pm 4.37\%$	19.76 ± 1.13	0.20 ± 0.04

2.3. Planning and Control System

Following the generation of personalized decision outputs, this study further designed a planning and control system deployable on in-vehicle domain controllers to enable real-time closed-loop operation of autonomous vehicles. For trajectory planning, polynomial curves are employed to generate smooth lane-change trajectories satisfying position, velocity, and acceleration constraints, while integrating driving styles and behavioral preferences to select optimal paths. For lateral control, a controller based on LQR and feedforward compensation was constructed to achieve stable regulation of lateral and yaw errors. For longitudinal control, a cascaded PID control structure was employed, ensuring longitudinal motion stability during lane changes through coordinated regulation of the position loop and velocity loop. This planning and control module precisely tracks the planned trajectory and seamlessly interfaces with the upper-level decision-making module, forming a complete autonomous driving control chain.

After comprehensively evaluating real-time requirements and the adaptability of path planning algorithms, a quintic polynomial curve was ultimately selected [27]. To simplify the trajectory planning process, it is assumed that longitudinal velocity remains constant during lane changes. The statistical analysis of the extracted trajectory dataset shows that the longitudinal velocity fluctuation during a single lane change maneuver is very small, often less than 0.1 m/s^2, proving this decoupling hypothesis with almost uniform motion.

Parameters for the lane-change trajectory function were determined based on road boundary constraints to reduce real-time computational load, yielding the following results:

$$\begin{cases} x(t) = v_x t \quad (D = v_x t) \\ y(t) = \frac{W}{T^5}(10T^2t^3 - 15Tt^4 + 6t^5) \end{cases} \tag{18}$$

where: T is the length of lane change; W is usually equal to the lane width and is taken as 3.75 m; and v_x is the longitudinal speed.

This paper adopts previous research: lateral LQR control and longitudinal cascade PID control [28]. First, a classical two-degree-of-freedom vehicle dynamics model is established:

$$ma_y = F_r + F_f \tag{19}$$

$$I_z\ddot{\delta} = L_f F_f cos\delta - L_r F_r \tag{20}$$

where V_f and V_r are the velocity vectors of the front and rear wheels of the vehicle, respectively; m is the mass of the vehicle, a_y is the longitudinal acceleration of the vehicle, and I_z is the moment of inertia. F_f is the front wheel lateral force, F_r is the rear wheel lateral force, L_f is the front wheel wheelbase, and L_r is the rear wheelbase. The lateral force F is given by $F = C\alpha$, where α is the tire lateral deflection angle. $C_{\alpha f}$ and $C_{\alpha r}$ denote the lateral deflection stiffness of the front and rear wheels, respectively. Given that the front wheel turning angle δ is typically small, $cos\delta \approx 1$. From the geometric relationships, it follows that:

$$\alpha_r = \frac{V_y - L_r\dot{\delta}}{V_x} \tag{21}$$

$$\alpha_f = \frac{V_y - L_f\dot{\delta}}{V_x} \tag{22}$$

where V_x is the vehicle velocity along the x-axis and V_y is the vehicle velocity along the y-axis. The three-degree-of-freedom vehicle dynamics model equations can be obtained after banding:

$$\dot{V}_y = -\frac{C_f + C_r}{mV_x}V_y + (-\frac{L_fC_f - L_rC_r}{mV_x} - V_x)\dot{\delta} + \frac{C_f}{m}\delta \tag{23}$$

$$\ddot{\delta} = -\frac{L_fC_f - L_rC_r}{I_zV_x}V_y + (-\frac{L_f^2C_f + L_r^2C_r}{I_zV_x})\dot{\delta} + \frac{L_fC_f}{I_z}\delta \tag{24}$$

The lateral position y and heading angle φ of the vehicle can be controlled by noting $\dot{x} = Ax + Bu$ by controlling the front wheel angle δ. Thus, the front wheel angle of the bicycle model can be converted to the steering wheel angle, and after calculation, the lateral control module in this study uses the steering wheel angle as the control input. The differential equation of the error is solved using Matlab's (R2022a, MathWorks, Natick, MA, USA) dlqr function, which first discretizes the error equation using Euler's method, and then iterates over the planned trajectory to find the closest trajectory point to the current position of the car, which is noted as the matching point, to obtain the error

$err = \left[ed, \ \dot{ed}, \ e\varphi, \ \dot{e\varphi}\right]$ for the discrete trajectory points. *ed* represents the lateral distance error, and $e\varphi$ the error of the vehicle direction angle φ and the tracking path angle φ_{des}.

The LQR algorithm requires defining a cost function that balances the cost of the system state (to ensure stability) and the cost of control effort (to minimize the amount of control applied). The selected cost function can be expressed as follows:

$$J = \frac{1}{2}\int_0^\infty \left(x^T Q x + u^T R u\right) dt \tag{25}$$

The final solution is obtained by solving the minimum cost function at the constraint $\dot{x} = (A - BK)x$:

$$u(t) = -Ke_{rr}(t) \tag{26}$$

where $\mathrm{K} = [\mathrm{k}_1, \mathrm{k}_2, \mathrm{k}_3, \mathrm{k}_4]$, these four control gains correspond to the four control state quantities in $\mathrm{e_{rr}}$, respectively. In order to eliminate the steady state error of the model, it is chosen to introduce the feedforward control δ_F, i.e., $\mathrm{u} = -\mathrm{Ke_{rr}} + \delta_\mathrm{F}$. After the introduction of the feedforward, $\dot{\mathrm{err}} = \mathrm{Aerr} + \mathrm{B}(-\mathrm{Kerr} + \delta_\mathrm{F}) + \mathrm{C}\dot{\varphi}_\mathrm{r}$, and when $\dot{\mathrm{err}} = 0$, the $\mathrm{err} = \mathrm{Aerr} + \mathrm{B}(-\mathrm{Kerr} + \delta_\mathrm{F}) + \mathrm{C}\dot{\varphi_\mathrm{des}}$, and to make $\mathrm{err} = 0$, δ_F is calculated as follows:

$$\delta_F = \frac{mv_x^2\rho}{L}\left(\frac{L_r}{C_f} - \frac{L_f}{C_r} + \frac{L_f}{C_r}k_3\right) - \rho(L_f + L_r - L_f k_3) \tag{27}$$

where ρ is the road curvature (curvature of the target point of the tracked path).

The longitudinal control module first applies PID control to the position error. The result is then combined with the desired speed and subjected to a second PID control based on the actual speed. Finally, this result is added to the desired acceleration to determine the required acceleration at the current moment. The acceleration to be performed at the current moment and the current actual speed can be compared with the throttle/brake calibration table to obtain the required brake pressure or throttle opening. Where the desired speed is v_{set} and the vehicle speed is v, the speed deviation can be:

$$e_{rr_{spe}} = v_{set} - v \tag{28}$$

Then the control law for maintaining uniform motion is:

$$u_{spe} = A \times e_{rr_{sep}} + B \times e_{rr_{spe_1}} + C \times e_{rr_{spe_2}} \tag{29}$$

where $A = k_p + k_i + k_d$, $B = -2k_i - k_p$, $C = \ k_d$; k_p, k_i, k_d are the three parameters of PID $e_{rr_{spe_1}}$ is the speed deviation of the last beat; $\mathrm{e_{rr_{spe_2}}}$ is the speed deviation of the last two beats. If there is a car ahead, a position loop needs to be introduced outside the velocity loop, and the position deviation is denoted as:

$$e_{rr_{dis}} = S_{set} - S \tag{30}$$

where $e_{rr_{dis}}$ is the position deviation, S_{set} is the set spacing, and S is the current head spacing. The control rate of the whole longitudinal control is:

$$u_{sum} = A\left(u_{dis} + e_{rr_{sep}}\right) + B\left(u_{dis} + e_{rr_{spe_1}}\right) + C\left(u_{dis} + e_{rr_{spe_2}}\right) \tag{31}$$

2.4. High-Fidelity Driving Simulation Platform

To support the drone data-driven personalized autonomous driving decision system for driving simulators, we developed an intelligent, connected-vehicle driving simulation platform. The platform uses an actual vehicle cockpit as the physical host and integrates a visual-scene and vehicle-dynamics simulation compute cluster, a host-PC control and network communication system, human–machine-interaction and actuator hardware, and drone-video injection and data-storage devices. Via Ethernet and in-vehicle bus connections, the platform achieves unified spatiotemporal synchronization and data exchange across a multi-layered "vehicle–road–person–air" simulation environment, thereby providing a stable and reliable operational basis for closed-loop testing of autonomous driving algorithms [29].

On the software side, the platform employs UC-win/Road (Version 16.0, FORUM8 Co., Ltd., Tokyo, Japan) and CarSim (Version 2016.1, Mechanical Simulation Corporation, Ann Arbor, MI, USA) to form the scenario simulation system: UC-win/Road is responsible for virtual-reality modeling of roads and traffic participants and for multi-channel scene rendering, while CarSim provides high-fidelity vehicle-dynamics models and responds to operating-condition parameters. In addition, user-defined plugins are implemented in C++ (Microsoft Visual Studio 2019, Microsoft Corp., Redmond, WA, USA) based on the UC-win/Road SDK to realize data interfaces and network communication between the simulation environment and external controllers. This architecture enables efficient construction of diverse traffic and test scenarios and supports online validation of personalized autonomous-driving decision algorithms. The overall platform appearance is shown in Figure 5.

Figure 5. Driving Simulation Platform.

For simulation software, this paper selected UC-win/Road [30], a virtual reality planning and design software for roads and landscapes. The scene simulation system primarily consists of UC-win/Road and CarSim: the former handles road scene construction, traffic participant animation rendering, and multi-channel display output, while the latter provides high-precision vehicle dynamics models and responds to operational parameters within the virtual scene. To meet the functional expansion and data interface requirements of the intelligent connected driving simulator, a user-defined plugin was developed in C++ based on UC-win/Road's underlying SDK. This plugin enables information processing and network communication between the simulation environment and external controllers. Leveraging UC-win/Road's comprehensive scene design and road editing capabilities, diverse traffic and test scenarios can be efficiently constructed. The simulation scene interface is shown in Figure 6.

Figure 6. Simulation Software Operation Diagram.

The parameter settings in Section 2 are intended to balance modeling fidelity and computational efficiency and to observe the robustness of the overall system performance to moderate variations in these parameters.

3. Results

3.1. Rationality Test

Based on the aerial trajectory data imported into the system, this study first verified the rationality of the lane-changing decision logic and control boundaries. The test environment was configured as a straight road section with a total length of 1000 m and a lane width of 3.75 m. The predefined scenario was as follows: the ego vehicle initially cruised at 80 km/h, switched to a car-following mode after detecting a slower vehicle ahead, executed a lane change once the lane-changing conditions were satisfied, and finally returned to the cruising state. The simulation duration was set to 20 s, with the lane-change maneuver triggered at approximately the 5th second and completed within 5 s. The timing of the lane-change maneuver was highly consistent with the behavior of real vehicles observed in the original aerial footage, while maintaining high yaw stability and lateral stability [31,32]. These results verify the effectiveness of the platform in reproducing real traffic behavior and executing personalized decision-making. The schematic trajectory is shown in Figure 7.

The kinematic differences among the three driving styles are illustrated in Figures 7–11. The aggressive style completes the lane change in approximately 3 s, with a peak lateral displacement of 0.8 m, whereas the normal and conservative styles require about 4 s and 5 s, respectively. The aggressive style also exhibits significantly larger dynamic variations, with longitudinal velocity fluctuations of about ± 0.2 m/s, a peak lateral velocity exceeding 0.4 m/s, and longitudinal acceleration of approximately ± 0.4 m/s^2, all of which are notably higher than those of the conservative style (with corresponding values below 0.3 m/s and ± 0.1 m/s^2).

In terms of attitude control, the peak yaw angle of the aggressive style approaches 0.4 rad, which is substantially greater than the 0.1 rad observed in the conservative style. Overall, the results reveal a clear and consistent pattern: the aggressive style is fast and abrupt, the conservative style is stable and gradual, and the normal style lies in between.

In addition, to further evaluate decision-making performance under complex interactions, three additional test scenarios are introduced.

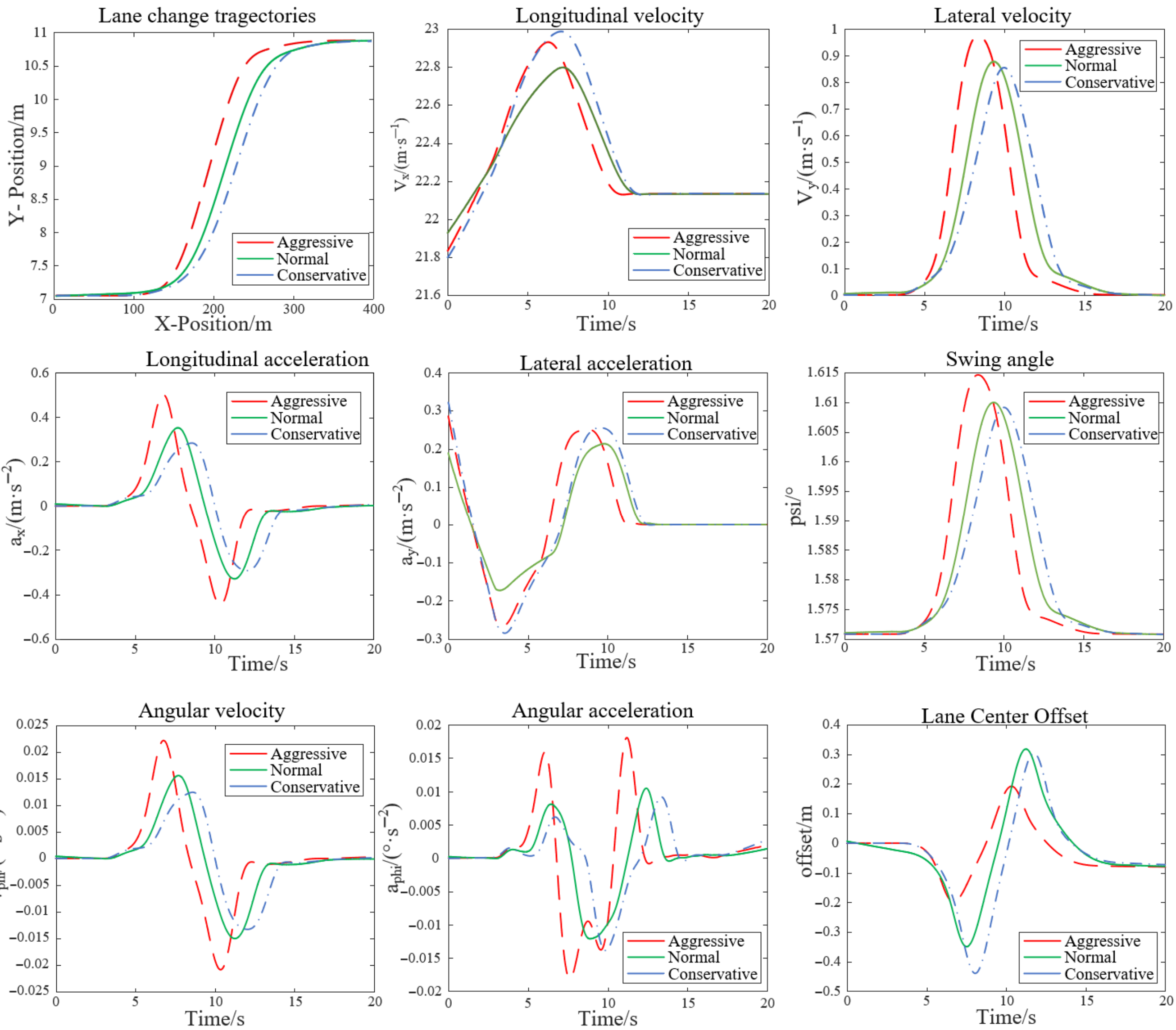

Figure 7. Lane Change Position and Speed Characteristics.

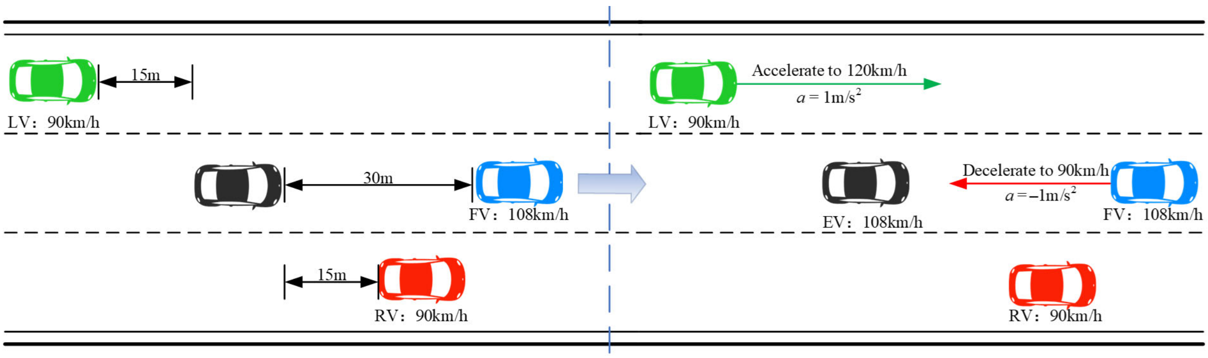

Figure 8. Front car deceleration scene. The dashed line is shown for visual reference and does not represent additional scientific information.

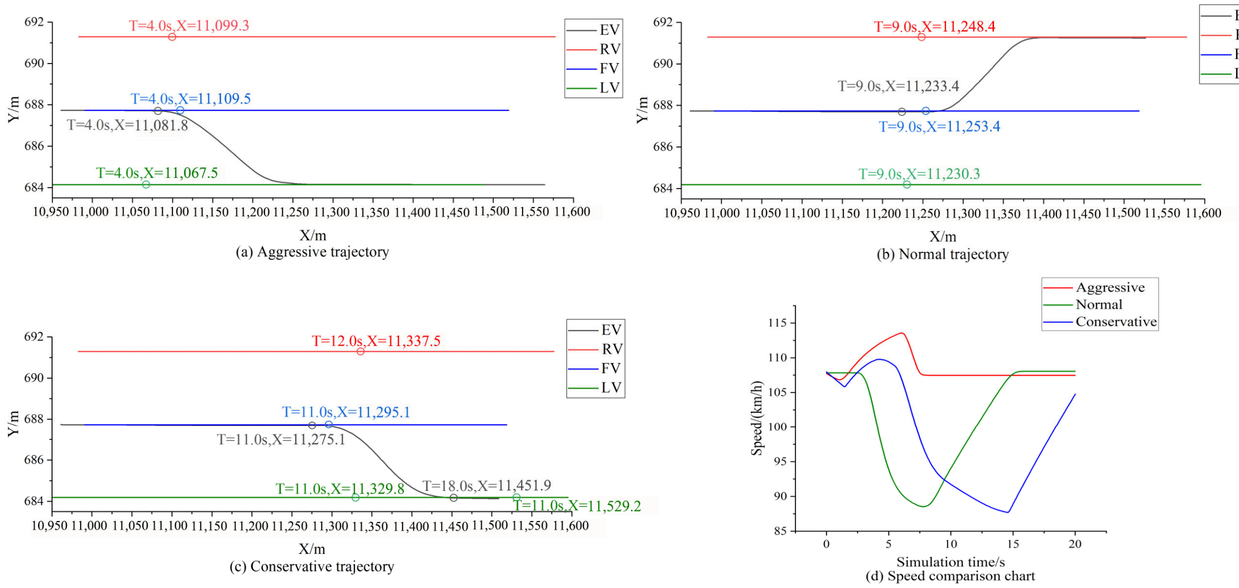

Figure 9. Comparison of driving parameters in three styles with the same scene.

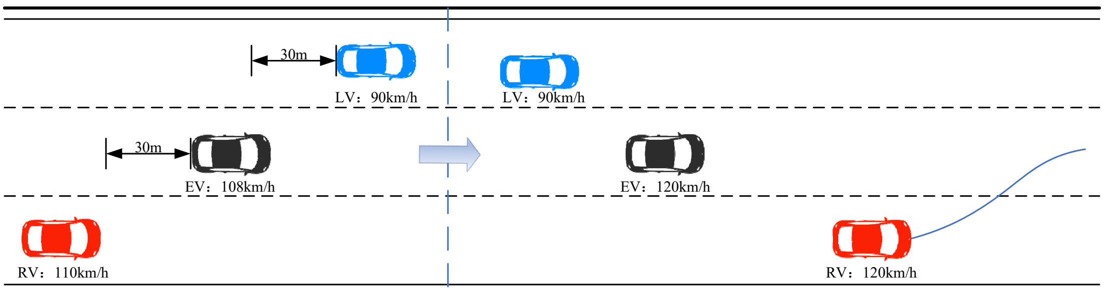

Figure 10. Right lane vehicle cuts into the scene. The dashed line is shown for visual reference and does not represent additional scientific information.

3.2. Driving Style Differentiation Scene

The deceleration scene of the vehicles ahead in the mixed traffic flow in Figure 8 validates the driving effect of the style preference of our research model on decision logic. Among them, EV represents the vehicle, FV represents the vehicle in front of the vehicle's lane, LV represents the vehicle of the left lane, and RV represents the vehicle of the right lane.

Set the acceleration of the vehicle on the left ($a = 1\ m/s^2$) and the deceleration of the vehicle ahead ($a = -1\ m/s^2$). The result is shown in Figure 9: Faced with the deceleration of the vehicle ahead, the attacking type chooses to accelerate quickly and use a small gap to charge left; The regular model adopts a deceleration and yielding strategy, with the left car changing lanes after passing through; The cautious type will significantly slow down and widen the distance before changing lanes to the right. This indicates that the model can generate differentiated strategies based on the trajectory predictions of surrounding vehicles, with a radical bias towards efficiency and a cautious priority for safety, effectively compensating for the limitations of traditional methods that rely solely on the current state.

3.3. Right Lane Vehicle Cuts into the Scene

Furthermore, to properly evaluate the value of personalization in highly interactive tasks, a dynamic scenario was tested where the ego vehicle must execute a left lane change to navigate between a faster-approaching vehicle on the right and a slower vehicle on the left (Figure 11). The proposed personalized model accurately identified the interaction context, initiating path planning early at T1 = 13.4 s and smoothly completing the maneuver at T3 = 26 s.

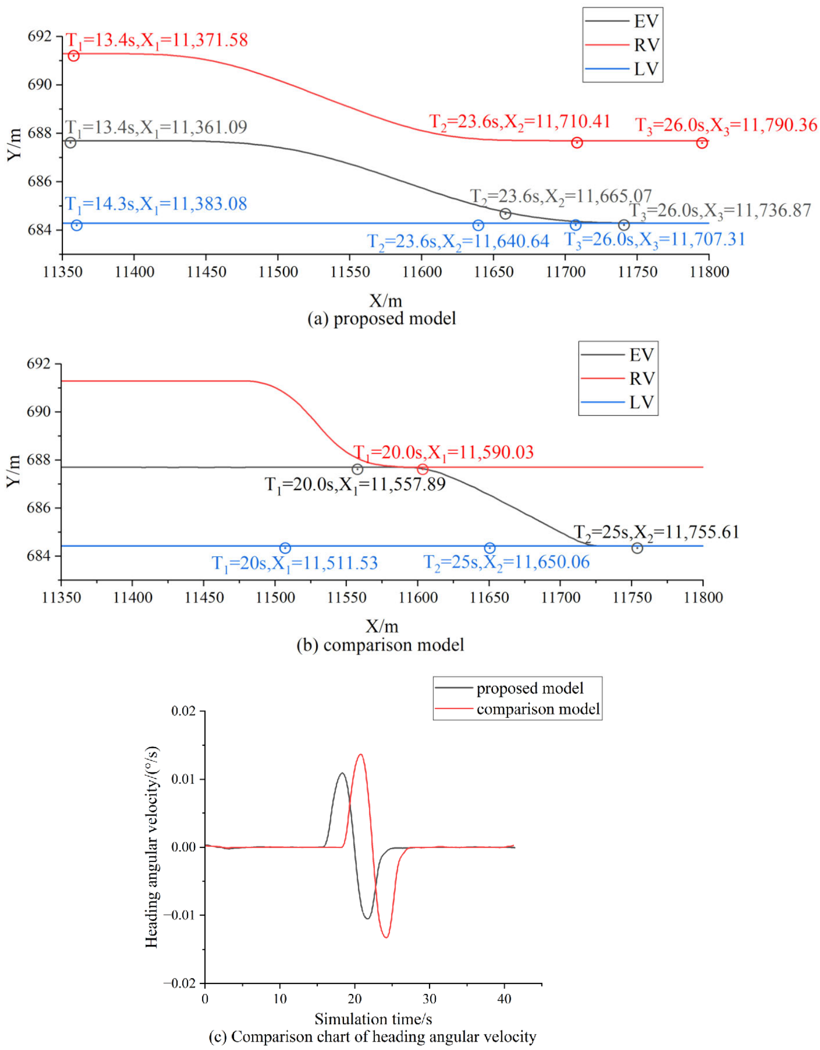

Figure 11. Comparison of driving parameters between two models in the same scenario.

In contrast, the conventional non-personalized baseline model, which is based on the classic Dynamic Programming and Quadratic Programming (DP + QP) framework, exhibited severe decision hesitation. Because the DP + QP model relies on unified, non-personalized cost functions and fixed safety margins, it struggles to adapt to complex multi-vehicle games. Consequently, it failed to initiate a timely lane change, resorting instead to passive deceleration at T1 = 20 s. The lane change was significantly delayed until T2 = 25 s, severely disrupting vehicle dynamic stability.

Quantitative analysis confirms that the personalized model maintained a lower heading angular velocity and achieved a significantly higher average Time-to-Collision (TTC) (18.04 s vs. 11.11 s). This vividly demonstrates that explicitly incorporating personalized driving styles effectively overcomes the rigid, non-adaptive nature of traditional non-personalized algorithms (like DP + QP), thereby simultaneously improving the safety margin and maneuver smoothness in dynamic environments.

4. Discussion

The results indicate that the proposed framework is capable of generating lane-changing behaviors that are not only dynamically feasible but also behaviorally differentiated, closely reflecting the heterogeneity observed in real-world traffic. Across all evaluated indicators, the aggressive, normal, and conservative driving styles exhibit systematic and internally consistent differences, suggesting that the learned policies capture intrinsic behavioral preferences rather than producing arbitrary variations in control outputs.

In terms of maneuver execution, aggressive driving is associated with earlier completion of lateral displacement, higher lateral velocities, and larger longitudinal speed fluctuations. These characteristics imply a shorter decision horizon and a stronger emphasis on efficiency, which aligns with established observations of aggressive human driving behavior. Conversely, conservative driving is characterized by prolonged lane-change durations and smoother kinematic profiles, indicating a preference for larger safety margins and reduced control intensity. The normal driving style consistently exhibits intermediate characteristics, supporting the interpretation of driving styles as a behavioral continuum rather than discrete categories.

The dynamic response analysis further reveals clear trade-offs between efficiency, comfort, and stability. Aggressive driving involves larger peaks in longitudinal and lateral acceleration, higher yaw rates, and greater angular accelerations, reflecting more abrupt control inputs and faster vehicle attitude changes. While such behavior facilitates rapid maneuver execution, it may increase discomfort and sensitivity to disturbances, particularly in dense or highly interactive traffic scenarios. In contrast, conservative driving maintains acceleration and attitude variations within narrower bounds, resulting in smoother trajectories and enhanced stability at the expense of responsiveness. The normal driving style achieves a balance between these competing objectives.

Despite the pronounced differences in driving behavior, all styles maintain acceptable yaw stability and lateral stability throughout the lane change process. This suggests that the underlying decision and control framework enforces fundamental vehicle dynamics constraints while allowing sufficient flexibility for behavioral differentiation. Moreover, the close correspondence between simulated lane-change timing and real vehicle behavior extracted from aerial photography data supports the realism of the constructed traffic scenario and the validity of the behavioral modeling approach.

From a broader perspective, these findings highlight the importance of incorporating driving style awareness into lane-changing decision-making for intelligent and automated vehicles. By explicitly modeling behavioral heterogeneity, the proposed approach enables more human-like interactions with surrounding traffic and provides a foundation for adaptive strategies that can respond to diverse driving contexts and user preferences. Such capabilities are particularly relevant for mixed traffic environments, where the coexistence of heterogeneous driving behaviors poses significant challenges to safety, efficiency, and user acceptance.

5. Conclusions

This study successfully designed and implemented a drone data–driven, personalized autonomous driving decision system for driving simulators. Through systematic experiments and validation, the platform met its intended objectives in driving-style modeling, personalized decision generation, and real-time planning and control. From a system perspective, the proposed approach can be interpreted as an integrated decision–planning–control architecture deployed and validated within a high-fidelity simulation environment. The specific conclusions are as follows:

(1) For driving style modeling, a natural driving database comprising 4997 real-world vehicle trajectories was constructed. Using an improved K-means++ clustering algorithm, drivers were categorised into three typical styles: aggressive, normal, and conservative. The clustering results revealed significant differences among the three styles in key metrics such as speed, acceleration, and headway distance, effectively

quantifying the behavioral characteristics of different driving styles and providing reliable data priors for personalized decision-making.

(2) For personalized decision-making, the proposed framework integrating dynamic games with maximum entropy inverse reinforcement learning (IRL) successfully learned personalized weightings for safety, comfort, and efficiency across driving styles from expert trajectories. For instance, the efficiency weighting for aggressive drivers reached 0.77, while the safety weighting for conservative drivers was 0.65, aligning with driving habits. Based on this, lane-change strategies generated using the DuDQN algorithm achieved over 85% accuracy compared to real expert decisions (90% for aggressive, 85% for normal, 87% for conservative). Additionally, validation in complex interactive scenarios confirms that the model effectively produces style-differentiated strategies—ranging from efficiency-oriented gap acceptance to safety-first yielding—by incorporating trajectory predictions, thereby overcoming the constraints of traditional rule-based approaches in dynamic mixed traffic.

(3) For planning and control, integrating quintic polynomial trajectory planning with lateral LQR feedforward and longitudinal cascade PID control algorithms achieves high precision, stable trajectory tracking. Experimental results show the vehicle maintains a maximum lateral tracking error below 0.1 m throughout lane changes, while longitudinal speed tracking error remains stable within ± 0.3 m/s. The dynamic responses of vehicles with different driving styles closely matched theoretical expectations: Aggressive driving exhibited the highest peak lateral acceleration and the shortest lane-change completion time, while conservative driving demonstrated the smoothest acceleration profiles and longer lane-change durations. This fully demonstrates the algorithm's effectiveness in reproducing the dynamic characteristics of diverse driving styles.

In summary, this platform not only establishes a complete technical chain from real-world data collection to in-vehicle controller deployment but also demonstrates, through extensive experimental data, its capability to faithfully reproduce natural traffic flows and generate personalized autonomous driving strategies highly consistent with human driving behavior. This research provides an efficient, reliable, and integrated solution for anthropomorphic validation and performance evaluation of autonomous driving algorithms. Future work will focus on validating the system's generalization capabilities in more complex urban scenarios and exploring collaborative decision-making mechanisms among multiple agents.

At the same time, it should be noted that several design choices in the framework, such as parameter settings and cost formulations, are based on empirical considerations to balance modeling fidelity and computational efficiency. Although the proposed system shows stable performance across different test scenarios, a more systematic sensitivity analysis and statistical evaluation (e.g., confidence intervals and broader sample validation) would further strengthen the robustness of the conclusions. Therefore, future work will focus on conducting comprehensive sensitivity studies, improving statistical validation, and extending the framework to more complex urban environments, including multi-agent interactive decision-making scenarios.

Author Contributions: Conceptualization, W.S.; methodology, W.S. and N.L.; software, Y.Z.; validation, W.S. and N.L.; formal analysis, W.S.; investigation, W.S. and Y.Z.; resources, W.S.; data curation, W.S. and N.L.; writing—original draft preparation, W.S.; writing—review and editing, W.S.; visualization, W.S. and Y.Z.; supervision, N.L.; project administration, W.S. All authors have read and agreed to the published version of the manuscript.

Funding: This work is supported by the Hubei Province Key Research and Development Program (2024BAB051), the Project of Hebei Expressway Group Co., Ltd. (Y01162511KY0001), the Natural Science Foundation of China (52472366), and the National Key Research and Development Program of China (2023YFB4302600).

Data Availability Statement: The data used in this study are derived from aerial trajectory data collected for research purposes. Due to privacy and usage restrictions, the data are not publicly available but can be provided by the authors upon reasonable request.

Conflicts of Interest: Author Wenpeng Sun was employed by the company Hebei Expressway Group Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ADAS	Advanced Driver Assistance Systems
AD-VILS	Automated Driving Vehicle-in-the-Loop Simulation
CSIRL	Curriculum Subgoal Inverse Reinforcement Learning
DuDQN	Dueling Network Architectures
LQR	Linear Quadratic Regulator
IRL	Inverse Reinforcement Learning
PID	Proportional–Integral–Derivative

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